

Quadratic Functions

LET – Mathematics and Statistics

$$y = ax^2 + bx + c$$

1. Solve the quadratic equations, which can all be factorised:

(a) $x^2 + 8x + 15 = 0$

(b) $x^2 + 2x - 15 = 0$

(c) $x^2 - 5x - 24 = 0$

(d) $x^2 + 6x = 0$

(e) $x^2 - 10x + 9 = 0$

(f) $x^2 + x = 30$

2. Solve the quadratic equations, which can all be factorised:

(a) $3x^2 + 7x + 4 = 0$

(b) $5x^2 + 12x + 4 = 0$

(c) $6x^2 + 7x = 5$

3. Solve the quadratic equations:

(a) $x^2 + 5x + 2 = 0$

(b) $x^2 + 7x - 1 = 0$

(c) $3 + 5x - x^2 = 0$

(d) $4x^2 - x = 1$

4. For each of the following quadratic functions, $y = f(x)$, find:

i The point where the function crosses the y -axis, i.e. $x = 0$

ii The points where the function crosses the x -axis, i.e. $y = 0$

iii The range of x , for which $f(x) > 0$

(a) $y = x^2 + 9x + 14$

(b) $y = x^2 - 3x + 2$

(c) $y = x^2 - 8x + 16$

(d) $y = 10 + 7x - x^2$

(e) $y = x^2 + 6x - 2$

(f) $y = 10 - 3x - 2x^2$